

enabled with both hard disks set as Master and Slave. Also, ensure that the correct hard disk drivers are installed, check your motherboard's drivers, and verify that they are of the latest version.

For optimum results, enable DMA mode—a tweak which has been described in the Tips & Tricks section of the magazine later on.

System Prerequisites

486 or higher CPU, 12 MB of RAM, 2 MB of hard disk space for the program

THE GRAPHICS SUBSYSTEM

For the avid 3D gamer in you, nothing less than a smooth 60 frames per second (fps) will do. There are a number of ways to go about checking out how fast your graphics subsystem performs.

Quake III

Though the technology behind the game was optimised for older generation graphics cards, *Quake III* is still the most widely used gaming benchmark.

To benchmark your graphics card using *Quake III*, load it up and wait until the main menu comes up. You must then proceed to the console window. Do so by typing the tilde character (~). Invoke the benchmarking process by first entering **timedemo 1** followed by the [Enter] key. On the next line, key in **demo demo001** and hit [Enter]. The game will load the demo and within moments, you will see a sequence of frames being played back at high speed. Sitting through the demo should only take a few seconds and it will end seemingly abruptly. When it does so, it will take you back to the main menu. To see your score, hit the tilde key again, and this will bring up the console window. Repeat this test at least two more times and take the average as your final score.

What to expect

500 MHz CPU with TNT2: 40 fps
1 GHz CPU with TNT2: 65 fps
1.5 GHz CPU with TNT2: 76 fps
(Tested at 1024x768 with 32-bit colour)

Warning

Before running this test, you might want to disable sound, which, if left enabled, could affect the scores you obtain in your graph-

ics tests. This could lead to a huge disparity—and make your graphics card look bad—when comparing scores across benchmarks, especially for those using



The lords of *Quake* don't want you for their next death-match, they want your video card!

built-in audio. Motherboards that come with built-in audio use soft audio, a technique that uses the CPU instead of audio hardware for processing sounds. This consumes lots of system resources and can cripple video performance as much as 20 per cent. You can disable on-board audio by entering the system BIOS and disabling it from there.

Benchmarks in *Quake III* are those that

shift heavily with changes in hardware. Even swapping out and exchanging motherboards can have an influence—by as much as 10 per cent. The key to keeping accuracy is to abide by some sort of uniformity in not only your hardware but also your software. We've seen that even disabling Windows Task Manager (with no tasks running) gave us a boost of about 2 fps. Of course, such a rise is negligible since most scores range in the 100s, but the point to note here is that the benchmark is very sensitive.

System Prerequisites

233 MHz processor, 32 MB RAM, 128 MB of free hard disk space, 3D graphics card with support for OpenGL

Quake III is perhaps the one video subsystem benchmark that has managed to gather a universal appeal for itself. Since it doesn't examine any specific features of a graphics card other than OpenGL 3D acceleration, it doesn't matter how old or how outdated your 3D card is.

Most users should be able to squeeze within these specs, though we strongly feel 32 MB of system memory to be seriously low. 128 MB is the sweet spot for the game; with so much memory, there could be no other bottlenecks, other than your video card, that is.

Serious Sam

For those of you who lack revolutionary gaming hardware yet don't subsist on what's left at the bottom of the technology barrel, there is plenty of middle ground. There are a couple of benchmarks for your video card that know what to check for, and how to check it.

Serious Sam is one game that features some pretty nice visuals, actually better than most games. And for this, it uses something known as a Transform and Lighting (T&L) engine. T&L is essentially a hardware feature ingrained into the core of a 3D accelerator card. Not all cards have this trait—the first one was nVidia's GeForce 256. All cards released after this one, ATI's Radeon for instance, have included support for this breakthrough in graphics technology.

The game uses OpenGL like *Quake III*, →

More Test Software

Quite a few system-wide inspection methods are available for download. Recently, MadOnion, the creators of 3DMark2001 SE, released a comprehensive diagnostic utility called PCMark 2002. It is an easy-to-use benchmark that delivers high-end diagnostic tools in the hands of novices. You can download this utility from www.madonion.com.

Sysmark 2002, made by BapCo, is another system-wide diagnostic tool that provides an application-based benchmark that reflects usage patterns for business users to enhance office productivity and aspects like design and multimedia. For more information, take a look at www.bapco.com.